

Arthur K. MacKeith

Contact information and digital portfolio at Arthur.Mackeith.info

EDUCATION

M.S., Computer Science, The University of Chicago

June 2020

Selected Course Work:

Math Foundations of Machine Learning, Introduction to Machine Learning, Algorithms, Computer Architecture, Parallel Programming, Time Series Analysis and Stochastic Processes

B.A., Physics, The University of Chicago

June 2020

Joint degree with M.S.

Cumulative GPA: 3.69/4.00,

Selected Course Work:

Honors Mechanics, Intermediate Mechanics, Mathematical Methods of Physics, Geophysical Fluid Dynamics I and II, Statistical Models and Methods, Statistical and Thermal Physics

Honors and Awards:

James Franck Institute Summer Research Fellowship (2018)

Dean's List (2016-2019)

PUBLICATIONS AND PRESENTATIONS

Lee, W-K., Preston, D. J., Nemitz, M. P., **MacKeith, A. K.**, Nagarkar, A., Gorissen, B., Vasios, N., Sanchez, V., Rawson, J., Bertoldi, K. Mahadevan, L., & Whitesides, G. M. A Robotic Bucklin-Sheet Ring Oscillator for Autonomous Locomotion. *Manuscript in Preparation*.

Murphy, K. A., **MacKeith, A. K.**, Roth, L. K., & Jaeger, H. M., (2019), The intertwined roles of particle shape and surface roughness in controlling the shear strength of a granular material. *Granular Matter*, 21(3). <https://doi.org/10.1007/s10035-019-0913-7>.

MacKeith, A. K., Murphy, K. A., Jaeger, H. M., 2019. Structure in Cylindrical Packings of Convex Particles. *University of Chicago Undergraduate Research Symposium*.

MacKeith, A. K., Lee, W-K, Whitesides, G. M.. 2019. Beam-Climbing Robot Controlled by a Soft Ring Oscillator. *Harvard University Campus Wide Poster Session*.

RESEARCH EXPERIENCE

Robotic Intelligence through Perception Lab, Toyota Technological Institute at Chicago, Chicago IL

Research Assistant

July 2020 to Present (Full time)
August 2019 to June 2020 (Part time)

- Jointly optimize the design and control of a soft robot through simulation
- Integrate a low-cost drone into education, outreach, and research robotics platform Duckietown

- Build a simulator for Duckietown drones to grade robotics assignments for high school and college students and benchmark novel algorithms in a model environment
- Improving hardware and software on the drone to effectively navigate the model town and communicate with the other vehicles

*George M. Whitesides Group, Harvard University,
Cambridge MA*

REU Student in the Harvard SEAS REU

June 2019 to August 2019

- Designed and built a cheap semi-soft beam climbing robot, incorporating soft controls such that it only requires a single constant input pressure gas source to climb
- Exploited static asymmetric friction to produce climbing from contracting actuators and optimized the asymmetry with the result that the robot was able to climb with a payload equal to its own weight (70 g), and perform common maintenance tasks like cleaning the beam as it climbed
- Ensured that the robot design used commercially available materials, with final cost below 5 USD
- Completed a pneumatic signal processing demonstration and characterized it using Matlab

*Jaeger Lab, The University of Chicago,
Chicago IL*

Research Assistant

December 2017 to June 2019

- Wrote and optimized image analysis python program that fit the position and orientation of the several thousand particles in a granular sample with 99.5% success rate
- Analyzed position and orientation data in python for global trends and mesoscale structures
- Designed 3D particles to print that break predictably to detect shifts in packing structure
- Recorded and reconstructed computed tomography scans of packings of granular media

OUTREACH

Physics with a Bang! Holiday Lecture and Open House, 2018

Volunteer,

- Guided groups of elementary, middle and high school students from the greater Chicago area through the Physics Department
- Explained hands on physics and chemistry demonstrations and lab equipment to students, families, and high school teachers

Duckietown (Mentioned above in Research Experience)

- Work to incorporate a low-cost raspberry pi based drone into the Duckietown platform
- Duckietown is a platform for education and research in robotics and AI and uses low cost components to demonstrate the principles of autonomy in the classroom. Both hardware and free learning materials are provided at Duckietown.org